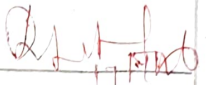


CO1 - ASSIGNMENT TEST - 1

SHORT ANSWER TEST

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Course Code : CSA0702
Course : Computer Networks
Faculty : Dr. R. S. Selvan
Batch : B TECH AI & DS
Date : 17/07/2026

Total Marks	20	Faculty signature	
Obtained Marks	19	Co-ordinator signature	

Transport and Data Link layers -

Given: File Size = 150 MB

Network Bandwidth = 50 Mbps

Transport layer segment size = 1500 bytes

Data Link Layer overhead = 40 bytes/frame

Assume 1 MB = 1,000,000 bytes.

Step: 1 Convert the file size into bytes:

$$\begin{aligned} 150 \text{ MB} &= 150 \times 1,000,000 \\ &= 150,000,000 \end{aligned}$$

Step: 2: Number of segments:

$$\frac{150,000,000}{1500} = 1,00,000$$

Since one segment is encapsulated into one frame,

Number of frames = 1,00,000

Step: 3 Data Link Layer overhead.

Overhead per frame = 40 bytes.

$$1,00,000 \times 40 = 40,00,000 \text{ bytes}$$

Total Overhead = 4 MB (or) 40,00,000 bytes.

Step: 4 Total Data Transmitted

$$15,000,000 + 40,000,000 = 15,400,000 \text{ bytes}$$

Convert to bits: 1 byte = 8 bits

$$15,400,000 \times 8 = 1,232,000,000 \text{ bits}$$

Step: 5 Transmission Time

$$\text{Bandwidth} = 50 \text{ Mbps} = 50,000,000 \text{ bits/s}$$

$$\frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds.}$$

Final Answer

- * Number of Transport Layer segments: 100,000
- * Number of Data Link Layer frames: 100,000
- * Total Data Link Layer Overhead: 4,000,000 bytes (4 MB)
- * Total Transmission time: 24.64 seconds.

Role of the Layers.

Transport Layer

- * Divides the file into 1500-byte segments.
- * Ensures reliable delivery using sequencing, acknowledgements, retransmissions, and error recovery.

Data Link Layer

- * Encapsulates each segment into a frame by adding 40-byte header and trailer.
- * Detects transmission errors using error-checking (e.g. CRC) and controls frame delivery between directly connected devices.

Conclusion:

The file is transmitted as 1,00,000 segments and 1,00,000 frames, with 4MB of Data Link Layer overhead requiring approximately 24.64 seconds to transmit over a 50 Mbps network (ignoring propagation delay).

2. Sliding Window Protocol

Given: Window size = 6 frames

Total frames = 48

Data / frame = 1024 bytes.

Calculation:

- * Number of transmission windows = $48 \div 6 = 8$ windows
- * One frame lost per window = 1 retransmission / window.
- * Total retransmission = $8 \times 1 = 8$ frames.

Answer

- * Transmission windows: 8
- * Total re-transmission: 8 frames.

Flow Control

- * Prevents sender from overwhelming the receiver.
- * Controls transmission rate using the sliding window.
- * Reduce congestion and improves communication efficiency.

Concept:

The Transport layer uses the sliding window protocol to send multiple frames before waiting for acknowledgment. This increases channel utilization and reduces idle time.

Low Control

- * Prevents the sender from overwhelming the receiver.
- * Matches transmission speed with receiver capacity.
- * Reduces congestion and packet loss.
- * Improves throughput and communication efficiency.

③ IP Fragmentation

Given: IP packet size = 12,000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Step: 1 Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Step: 2 Number of fragments

$$\frac{12000}{1480} = 8.11$$

Round-off = 9 fragments

Step: 3 Header overhead:

$$9 \times 20 = 180 \text{ bytes}$$

The Network layer fragments packets when they exceed the MTU.

Every fragment contains its own IP header with.

- (i) Identification field
- (ii) Fragment Offset
- (iii) More Fragment MF flag.

The destination host reassembles all fragments into the original packet.

Answer : Fragments = 9

Header Overhead = 180 bytes.

④ Sliding Window Protocol

Given:

Sliding Window size = 10 frames

Data per frame = 1024 bytes

Total frames = 48

Step 1. Number of transmission windows

$$\frac{48}{12} = 4 \text{ windows}$$

Step 2 Retransmission

* One frame is lost in each window

* Number of windows = 4

Total Retransmission = 4 frames

The transport layer uses the sliding window protocol to send multiple frames before waiting for acknowledgements. This increases channel utilization and reduces idle time.

Flow Control

* Prevents the sender from overwhelming the receiver.

* Matches transmission speed with receiver capacity.

* Reduces congestion and packet loss.

* Improves throughput and communication efficiency.

Answer

* Transmission windows = 4

* Retransmission = 4 frames

5. Session Layer Synchronization

Given: Video File = 600MB

Checkpoint every 60MB

Failure after 390MB

Step 1: Number of checkpoints

$$\frac{390}{60} = 6.5$$

Completed checkpoints: 6 checkpoints

(at 60, 120, 180, 240, 300, 360 MB)

Step 2: Retransmission.

Failure occurred at 390 MB

Last checkpoint = 360 MB

Retransmission required = $390 - 360 = 30$ MB

The Session Layer inserts synchronization points (checkpoints)

After a communication failure, transmission resumes from the last checkpoint instead of restarting from the beginning.

Answer: Checkpoints = 6

Retransmission = 30 MB

Present Layer

Given: * Original file = 900 MB

* Compression = 40%

* Encryption overhead = 5%

Step: 1 Compressed size

40% Reduction

$$900 \times 0.60 = 540 \text{ MB}$$

Step: 2 Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Step: 3 Percentage Reduction

$$900 - 567 = 333 \text{ MB}$$

$$\text{Percentage} = \frac{333}{900} \times 100 = 37\%$$

10. Presentation Layer

- * Compresses data to reduce bandwidth usage.
- * Encrypts data for confidentiality.
- * Converts data format for compatibility between systems.

Answers:

* Compressed size = 540 MB

* Encrypted size = 567 MB

* Percentage reduction = 37%

④ FTF Application

Given

* Data = 3 GB

* Throughput = 120 Mbps

* Request size = 3 MB

Assume: 1 GB = 1000 MB

Total data: 3 GB = 3000 MB

Step: 1 Requests

$$\frac{3000}{3} = 1000$$

Step: 2 Transmission Time

Converts to bits:

$$3000 \text{ MB} \times 8 = 24000 \text{ Mb}$$

$$\frac{24000}{120} = 200 \text{ seconds}$$

The Application Layer provides services like FTP, HTTP and Email. It ensures user authentication, file transfer, naming services and application-level communication.

Answer: Requests = 1000

Transmission Time = 200 seconds.

nd to End Delay

Processing delays.

Application = 4ms

Transport = 3ms

Network = 5ms

Data link = 2ms

Physical = 1ms

Total processing: $4 + 3 + 5 + 2 + 1 = 15\text{ms}$

Propagation: 18ms

Transmission: 12ms

Total delay: $15 + 18 + 12 = 45\text{ms}$

Every OSI layer performs a different function.

* Application: User services

* Transport: Reliable delivery.

* Network: Routing

* Data Link: Framing and error detection

* Physical: Bit transmission.

Answer: Total end-to-end delay = 45ms

7. Protocol Overhead

- Given :
- * Useful data = 80 MB
 - * Frame size = 1600 bytes
 - * Overhead = 80 bytes.

Useful payload: $1600 - 80 = 1520$ bytes

Useful data : 80 MB = 80,000,000 bytes

Step 1: Frames: $\frac{80,000,000}{1520} = 52631.58$

Frames: 52,632

Step 2: Total Overhead

$$52632 \times 80 = 4,210,560 \text{ bytes} = 4.21 \text{ MB}$$

Step 3: Efficiency:

$$\frac{1520}{1600} \times 100 = 95\%$$

Protocol overhead contains addressing, sequencing, error checking and control information. While it reduces efficiency slightly it enables reliable and secure communication.

Answer

* Frames = 52,632

* Overhead = 4,210,560 bytes $\approx$ 4.21 MB

* Efficiency = 95%

Hospital Medical Record.

Given * Original file = 240 MB

* Compressor = 30%

* Segment size = 1400 bytes

* Data Link Overhead = 60 bytes

Step: 1 Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

$$\text{Convert } 168 \text{ MB} = 168,000,000 \text{ bytes}$$

$$\text{Step: 2 Number of segments} = \frac{168,000,000}{1400} = 120,000$$

$$\text{Step: 3 Protocol overhead} : 120,000 \times 60 = 7,200,000 \text{ bytes} \\ = 7.2 \text{ MB}$$

$$\text{Step: 4 Final Transmitted data} : 168 \text{ MB} + 7.2 \text{ MB} = 175.2 \text{ MB}$$

Multiple OSI layer cooperate during transmission.

- * ~~Presentation~~ Layer: Compresses and encrypts the patient's medical record for efficiency and security.
- * Transport Layer: Divides the compressed data into 1400 byte segment and ensures reliable delivery using sequencing, acknowledgement and retransmission.
- * Data Link Layer: Adds a 60-byte header and trailer to each segment for framing and error detection.
- * Physical Layer: Transmits the final bit stream across the communication medium.
- * Together these layers provides secure, accurate and reliable transfer of sensitive healthcare information.

Criteria	Weightage	
Understanding of Concepts	25-1.	5
Explanation	25-1.	5
Application	20-1.	4
Organization	15-1.	3
Accuracy	10-1.	1
Clarity	5-1.	1